

Bahria University,

Karachi Campus



COURSE: Data Mining

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PROJECT NAME: Cars Classification Model

Engr.Aamana

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Car Classification Model Report

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1. Introduction & Problem

The objective of this project is to develop a robust car classification model using machine learning and deep learning techniques. The classification model aims to identify different types of Toyota cars based on input images. This system is intended to streamline tasks such as inventory management, search optimization, and targeted recommendations in automotive sales and service industries.

Problem Statement

Traditional methods for identifying car models are manual and prone to errors. With the growth of online marketplaces and automated systems, accurate classification is essential for seamless integration. This project addresses this gap by leveraging machine learning.

2. Technology

Backend

- Python: For building the model and image preprocessing.
- TensorFlow: Deep learning framework used to create and train the convolutional neural network (CNN) for image classification.
- NumPy: For numerical computations.
- OpenCV: For image manipulation and preprocessing.
- Matplotlib & Seaborn: For data visualization and evaluation metrics.

Frontend

- React.js: Implemented for user interface design and displaying results.

Libraries Used

- tensorflow and tensorflow-gpu: To build and optimize the CNN model.
 - opencv-python: Image reading and processing.
 - matplotlib: Visualizing data and model performance.
 - seaborn: Plotting the confusion matrix.
 - Pillow (PIL): Verifying image files.
 - imghdr: To validate image formats.
 - sklearn: For evaluation metrics like confusion matrix and classification report.
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3. Functionalities

Data Validation:

Ensures input images are valid and removes corrupt or unsupported files.

Data Augmentation:

Techniques like zooming, width/height shifting, and normalization improve model generalization.

Model Training:

Trains a CNN for car classification using training and validation datasets.

Evaluation:

Evaluates performance using metrics like accuracy, confusion matrix, and classification report.

Prediction:

Displays predictions for test images along with visualizations.

Recommendation System:

Suggests car models to users based on their preferences and input.

4. Module Distribution

Data Preprocessing Module:

- Reads images and validates formats.
- Augments data for better generalization.

Model Building Module:

- Defines a sequential CNN architecture.
- Includes layers such as convolution, max-pooling, and dropout.

Training Module:

- Compiles the model with Adam optimizer and categorical crossentropy loss.
- Implements callbacks like ReduceLROnPlateau for dynamic learning rate adjustment.

Evaluation Module:

- Generates accuracy and loss metrics.
- Creates confusion matrices and visualizations for better insights.

Frontend Integration Module:

- Displays results via a React.js interface.

5. Code

The project code includes the following key sections:

Data Preprocessing

```
from PIL import Image
import imghdr

def validate_images(dataset_path):
    invalid_files = []
    for root, _, files in os.walk(dataset_path):
        for file in files:
            file_path = os.path.join(root, file)
            try:
                with Image.open(file_path) as img:
                    img.verify()
            except Exception as e:
                invalid_files.append(file_path)
    return invalid_files
```

Model Architecture

```
def create_model():
    model = Sequential([
        Conv2D(128, (5, 5), activation='relu', input_shape=(150, 150, 3)),
        MaxPooling2D(2, 2),
        BatchNormalization(),

        Conv2D(64, (3, 3), activation='relu'),
        MaxPooling2D(2, 2),
        BatchNormalization(),
```

```
Conv2D(32, (3, 3), activation='relu'),
MaxPooling2D(2, 2),
BatchNormalization(),

Flatten(),
Dense(256, activation='relu'),
Dropout(0.5),
Dense(5, activation='softmax')
])
return model
```

Training

```
cnn_model.compile(optimizer=Adam(), loss=CategoricalCrossentropy(), metrics=['accuracy'])
cnn_model.fit(train_generator, epochs=10, validation_data=validation_generator, callbacks=[reduce_lr])
```

Evaluation

```
cf_mtx = confusion_matrix(y_true, y_pred)
sns.heatmap(cf_mtx, annot=True, cmap="YlGnBu", xticklabels=labels.values(), yticklabels=labels.values())
plt.show()
```

Front End:

[RizwanAnsar2004/Car-Classification](#)

6. Interfaces

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CLASSIFY YOUR CAR

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File should be an image

Drop files here

or

Choose File

No file chosen

SUBMIT

7. Conclusion

The car classification model effectively identifies Toyota car models with high accuracy. The inclusion of a recommendation system enhances user experience, making the tool suitable for e-commerce and inventory management. Future improvements include expanding the dataset and integrating more car brands.